

A. Count the Inversion

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: `sort`, `open`, `file`, `creat`(, `fstream`, `thread`, `process`, `system`(, `exec`(

Here is a Pseudocode of the Merge Sort Algorithm.

```
def merge(a, b):
    # write your code here
    # a and b are two sorted List
    # merge function will return a sorted List after merging a and b

def mergeSort(arr):
    if len(arr) <= 1:
        return arr
    else:
        mid = len(arr)//2
        a1 = mergeSort(.....) # write the parameter
        a2 = mergeSort(.....) # write the parameter
        return merge(a1, a2) # complete the merge function above
```

Now, you are given an array A of size N of N distinct integers. It is guaranteed that the array A contains a permutation of integers from 1 to N (i.e., every integer from 1 to N appears exactly once).

1. Count the number of inversions in the given array.
2. Sort the array in non-decreasing order.

An inversion is a pair (i, j) where $i < j$ and $A[i] > A[j]$.

Input

The first line contains an integer N ($1 \leq N \leq 10^5$) — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq N$) separated by spaces.

Output

In the first line, print the total number of inversions in the given array. In the next line, print the array in non-decreasing order.

Examples

input	Copy
5 1 2 5 4 3	
output	Copy
3 1 2 3 4 5	

CSE221 Assignment 03 Summer 2026

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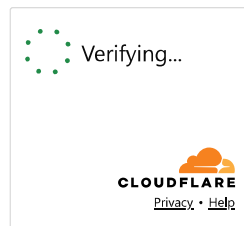
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input	Copy
5 1 2 3 4 5	
output	Copy
0 1 2 3 4 5	

input	Copy
5 5 4 3 2 1	
output	Copy
10 1 2 3 4 5	

input	Copy
7 6 4 2 5 7 3 1	
output	Copy
14 1 2 3 4 5 6 7	

Note

In the first example (1-based indexing), the inversions are the pairs of indices (3, 4), (3, 5) and (4, 5).

In the second example, there are no inversions.

In the third example, every pair of i, j where $i < j$, we have $A[i] > A[j]$. Hence, All 10 such pairs are inversions.

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